

The empirical recurrence for OEIS A300494, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A300494 records “Empirical recurrence of order 83” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 107$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{83} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-83 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A300494 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 2 or 4 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 50, 143, 499, 1513, 4686, 14822, 46717, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Mar 07 2018 07:22:07, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 107$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 107$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 83: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 83.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 214$ exact terms from $n = 4$ on, it returns order 83 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	11	c_{22}	−41650	c_{43}	6457172	c_{64}	−437989
c_2	−51	c_{23}	−25768	c_{44}	5036998	c_{65}	−7528
c_3	145	c_{24}	32651	c_{45}	1301071	c_{66}	160459
c_4	−356	c_{25}	−138394	c_{46}	−2987469	c_{67}	189913
c_5	903	c_{26}	129824	c_{47}	−5621412	c_{68}	127773
c_6	−2013	c_{27}	42450	c_{48}	−5498876	c_{69}	48223
c_7	3662	c_{28}	293882	c_{49}	−3517590	c_{70}	1748
c_8	−6416	c_{29}	581525	c_{50}	−1323137	c_{71}	−14162
c_9	11278	c_{30}	231216	c_{51}	−744933	c_{72}	−11280
c_{10}	−16413	c_{31}	180236	c_{52}	−1928150	c_{73}	−4405
c_{11}	19852	c_{32}	−883168	c_{53}	−2550613	c_{74}	−386
c_{12}	−24515	c_{33}	−1402510	c_{54}	−1454436	c_{75}	948
c_{13}	25721	c_{34}	−1198904	c_{55}	852850	c_{76}	570
c_{14}	−12929	c_{35}	−844078	c_{56}	3013037	c_{77}	69
c_{15}	125	c_{36}	242904	c_{57}	3634442	c_{78}	−75
c_{16}	9514	c_{37}	292028	c_{58}	2964991	c_{79}	−74
c_{17}	−35395	c_{38}	−597944	c_{59}	1653123	c_{80}	−31
c_{18}	39169	c_{39}	−928724	c_{60}	75245	c_{81}	0
c_{19}	−13646	c_{40}	−396157	c_{61}	−1058211	c_{82}	4
c_{20}	4510	c_{41}	2026012	c_{62}	−1404974	c_{83}	1
c_{21}	15646	c_{42}	5107888	c_{63}	−1067074		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A300494 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{83} - c_1x^{83-1} - \dots - c_{83}$. Its constant term is $-c_{83} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 83 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 83, so $\deg q = 83$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 83, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 83, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-83 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A300494.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.